# Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

## Lesson and revision-note context

| | |
|---|---|
| **Subject** | Mathematics |
| **Grade** | JSS1 |
| **Term** | Third Term |
| **Week** | 9 |
| **Topic** | Need for statistic |
| **Subtopic** | Need for statistics: purposes, and the collection and tabulation of data |
| **Theme** | — |
| **Curriculum objectives** | • explain the purpose of statistic<br>• analyse and interpret statistical data presented in tables |
| **Previous knowledge** | • Students are familiar with counting and recording information. |
| **Lesson sequence** | • explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information<br>• analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table |
| **Instructional materials** | • classroom register, bottle tops, pencils or counters. |
| **Evaluation** | • (a) Define statistic<br>• (b) State three methods of collecting data.<br>• (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2. |

## Lesson objectives

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|---|---|
| **1.** | explain the purpose of statistic |
| **2.** | analyse and interpret statistical data presented in tables |

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

# STUDENT REVISION NOTE

**Topic:** Need for statistics: purposes, and the collection and tabulation of data

**Class:** JSS1 **Subject:** Mathematics **Week:** 9

## Section 1: Why We Use Statistics

Every day you are surrounded by numbers. The number of students in your class, the scores in a test, the number of bottles in a crate, the ages of people in your family — all of these are pieces of information. On their own, scattered here and there, they are not very useful. But when you collect them, organise them and make sense of them, they become powerful. That is what statistics is about.

Statistics is the method of collecting and organising information so that it can be understood and used. It helps us answer questions like: How many? How many more? How many altogether? What is the most common?

Why do we need statistics? Because raw information, on its own, tells us very little. Once we collect it and put it in order, we can see what is happening, compare things, and make good decisions. Suppose a shopkeeper wants to know which drink sells best. If he just watches customers, he might guess. But if he counts and organises the sales in a table, he knows for certain which one to order more of. That is the purpose of statistics: to turn scattered information into something clear and useful.

Statistics is made of two main actions: collecting information and organising it.

Collecting means gathering the information from where it is. You might count the number of girls and boys in your class, or ask everyone their favourite colour.

Organising means putting the collected information in a neat, ordered form, usually in a table, so that it is easy to read.

Watch how collecting and organising work together. A teacher wants to know how many students are in each of two arms of Class 1. She counts: 24 students in 1A and 27 students in 1B. She has collected the information.

Now she organises it in a table:

| Class | Number of students |
|---|---|
| 1A | 24 |
| 1B | 27 |

Now the information is clear at a glance. You can see at once that 1B has more students than 1A, and you can add to find that together they have $24 + 27 = 51$ students. That is the whole point of statistics — collecting the facts and organising them so the answers become obvious.

Let's work through collecting and organising information together.

<u>Example.</u> A group of friends wants to know how many red, blue and green pencils they have altogether. Amina counts and finds 8 red pencils, 5 blue pencils and 7 green pencils.

First, the information is collected: Amina has counted 8 red, 5 blue and 7 green.

Next, it is organised in a table so that it is easy to read:

| Colour of pencil | Number of pencils |
|---|---|
| Red | 8 |
| Blue | 5 |
| Green | 7 |

Now the organised table lets us answer questions easily. How many pencils are there altogether? Add the column:

$8 + 5 + 7 = 20$

So there are 20 pencils in total. The table also shows that red pencils are the most common (8) and blue pencils are the fewest (5). This is exactly why we collect and organise information: scattered counts become a clear picture.

**Practice**

1. Define statistics in your own words.
2. A fruit seller counts the fruit she has: 15 oranges, 12 mangoes and 9 pawpaws. Organise this information in a table.
3. Using your table from the previous question, state how many fruits the seller has altogether, and name which fruit is the fewest.

## Section 2: Organising Data with Tally Marks and Frequency Tables

Raw data on its own is hard to read. If you are asked how many of your classmates are 12 years old, you do not want to count through a long, jumbled list every time. So we organise data into a table. Two tools help us do that: tally marks and a frequency table.

Tally marks are a quick way to count how many times each value appears. You draw one stroke for each item. When you reach the fifth, you draw it across the four like this: 𝍸 (four strokes, then the fifth crossing them). That makes groups of five, so you can count them in fives at the end.

A frequency table sets out each value together with the number of times it appears. The table has columns for the value, the tally marks, and the frequency (the count). Once the table is built, you can answer questions about the data just by reading the table.

Worked example. A class was asked how many pets each student owns, and these were the answers:

3, 1, 2, 0, 2, 3, 1, 2, 0, 3, 2, 1, 2

Let us build a frequency table.

Step 1 — List the values. The values that appear are 0, 1, 2 and 3. Write them in a column in order, smallest first.

Step 2 — Tally. Go through the list one answer at a time and make a tally mark beside that value:
- 0 appears twice: 𝍷
- 1 appears three times: 𝍷 (three strokes)
- 2 appears five times: 𝍸
- 3 appears three times: 𝍷

Step 3 — Count the tally marks. Remember that a crossed group 𝍸 counts as five. So the frequencies are: 0 has 2, 1 has 3, 2 has 5, 3 has 3.

Step 4 — Write the frequency table.

| Number of pets | Tally | Frequency |
|---|---|---|
| 0 | 𝍷 | 2 |
| 1 | 𝍷 | 3 |
| 2 | 𝍸 | 5 |
| 3 | 𝍷 | 3 |

Now we can read answers straight from the table. How many students own 2 pets? The frequency for 2 is 5. How many students answered altogether? Add the frequencies: 2 + 3 + 5 + 3 = 13 students.

**Practice**

1. Prepare a frequency table for the following data:

   1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2
2. A frequency table for the ages of students in a club is shown below.

| Age | Tally | Frequency |
|---|---|---|
| 11 | 𝍸 | 5 |
| 12 | 𝍸 𝍷 | 7 |

| Age | Tally | Frequency |
|---|---|---|
| 13 | [illegible] | 6 |
| 14 | [illegible] | 3 |

(a) How many students are 12 years old?
(b) Which age has the highest frequency?
(c) How many students are in the club altogether?

# Check your work

Attempt every question before you read this page.

## Why We Use Statistics

1. Statistics is the method of collecting and organising information so that it can be understood and used.
2. | Fruit | Number |
|---|---|
| Oranges | 15 |
| Mangoes | 12 |
| Pawpaws | 9 |

   Collect the counts: 15 oranges, 12 mangoes, 9 pawpaws. Organise them into a table with one row for each fruit and a column showing the count.
3. Altogether there are 15 + 12 + 9 = 36 fruits. The pawpaws, with 9, are the fewest.

   Add the organised counts: 15 + 12 + 9 = 36. Compare the counts in the table: 9 is the smallest, so pawpaws are the fewest.

## Organising Data with Tally Marks and Frequency Tables

1. The frequency table is:

   | Number | Tally | Frequency |
|---|---|---|
| 1 | [illegible] | 3 |
| 2 | [illegible] | 3 |
| 3 | [illegible] | 4 |
| 4 | [illegible] | 3 |

   Total = 13

   List the values in order: 1, 2, 3, 4. Go through the data and tally each occurrence. 1 appears three times, 2 appears three times, 3 appears four times (a crossed group of five is not needed since four is fewer than five), and 4 appears three times. Add the frequencies to check: 3 + 3 + 4 + 3 = 13, which matches the 13 numbers in the data.
2. (a) 7 students
   (b) Age 12 (frequency 7)
   (c) 5 + 7 + 6 + 3 = 21 students

## Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

| | | | |
|---|---|---|---|
| **Revision note ID** | cand-d0ff05c1 | **Date** | |
| **Subtopic** | Need for statistics: purposes, and the collection and tabulation of data | **Rater** | |

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

| **Criterion** (select one score per row) | **1** Poor | **2** Weak | **3** Adequate | **4** Good | **5** Excellent |
|---|---|---|---|---|---|
| **Objective and curriculum alignment** Does the revision note cover the intended lesson objectives and curriculum target? | ○ | ○ | ○ | ○ | ○ |
| **Taught-lesson alignment** Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth? | ○ | ○ | ○ | ○ | ○ |
| **Scope and conceptual correctness** Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors? | ○ | ○ | ○ | ○ | ○ |
| **Student-facing clarity and usefulness** Can JSS1 learners understand and use it meaningfully as take-home review material? | ○ | ○ | ○ | ○ | ○ |

| Strengths | Concerns | Recommendation |
|---|---|---|
| | | |

## Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

| | |
|---|---|
| ☐ | **Misses a required objective** — Skips something the lesson was meant to teach. |
| ☐ | **Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic). |
| ☐ | **Wrong or misleading content** — Contains a mathematical or factual mistake. |
| ☐ | **Too hard for JSS1 to follow** — Likely confusing for the students. |
| ☐ | **Other:** _____ |